

Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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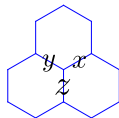


1 Kitaev Model

Kitaev Model: an Introduction

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ;$$



- Majorana fermion (parton) representation: $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = ib^\mu c$
- Parton Hamiltonian

$$H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left(\frac{1 + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL105 080501)

$$S_A = S_A^F + S_A^G$$

Kitaev Model: an Introduction